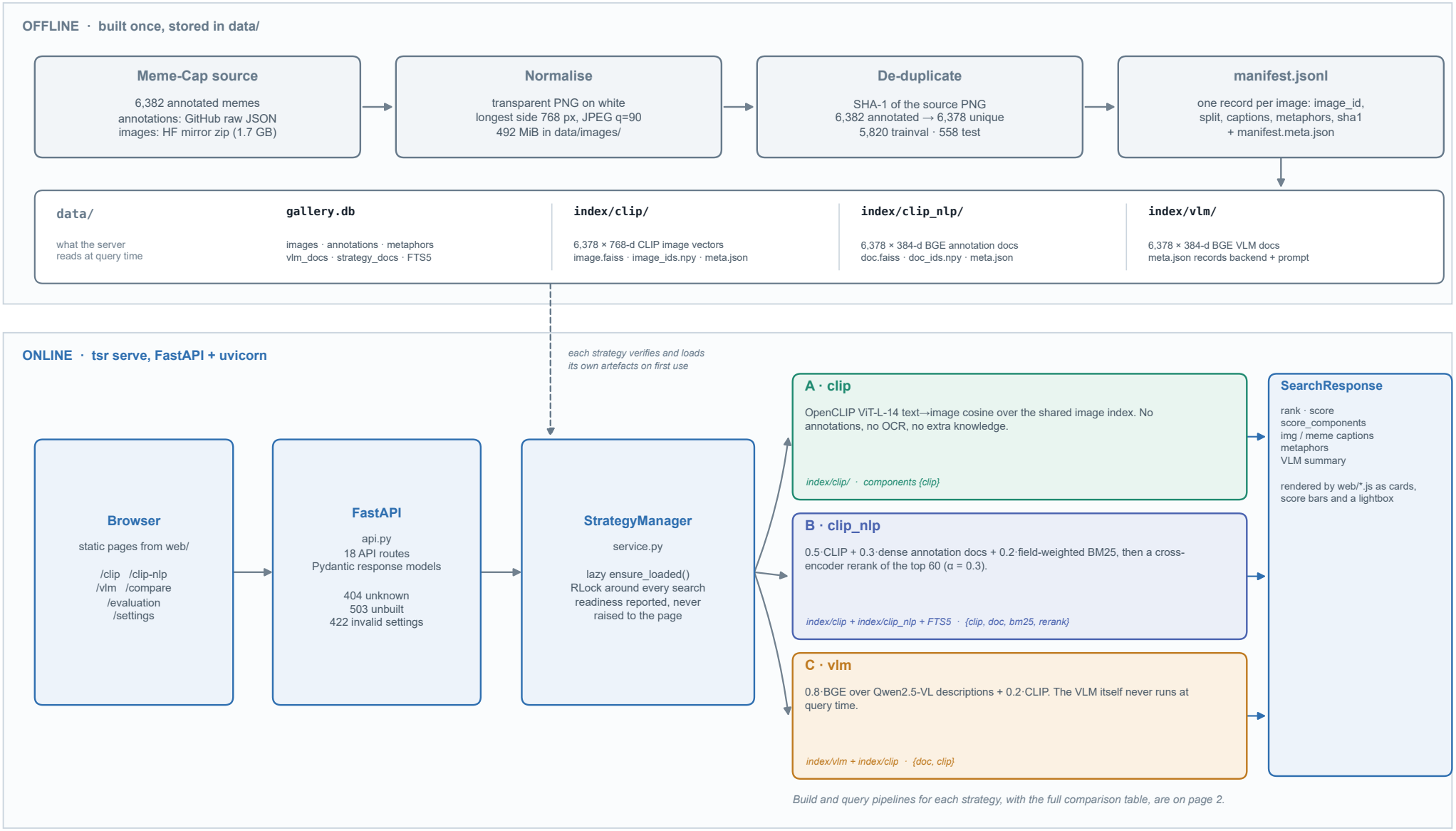
System architecture

One Meme-Cap gallery, three representations of the same 6,378 images, one benchmark and one FastAPI service



Build and query pipelines for each strategy, with the full comparison table, are on page 2.

INVARIANTS images.id is the only key shared by SQLite, FAISS row ids and image_ids.npy — assigned once, never renumbered. Every index carries meta.json (schema, model, dim, count, dataset hash); a mismatch is refused, never served.

Measured on the frozen set (240 queries): mAP A 0.4395 · B 0.7608 · C 0.4519. Artifact counts from data/index/*/meta.json.

The three strategies, side by side

Same gallery, same CLIP image vectors, same API — the only difference is which text, if any, represents an image at query time.

A · clip

Pixels only — the zero-knowledge baseline

BUILD · once, offline

Encode the gallery
CLIP ViT-L-14 image tower
6,378 vectors, L2-normalised

Store the index
image.faiss + image_ids.npy
meta.json records model and dim

`index/clip/ · 768-d`

QUERY · every request

Encode the query
CLIP text tower, 77 tokens
no annotations are ever read

Inner product
against 6,378 image vectors
inner product = cosine similarity

Cut the list
drop every score ≤ 0
keep the top k

Sees the scene, not the joke: overlaid text is invisible to it.

B · clip_nlp

Human annotations, fused with the pixels

BUILD · once, offline

Four fields per image
scene · meaning · metaphor · title
1,968 literal metaphor rows dropped

Write two stores
dense doc → BGE-small, 384-d
same fields → FTS5 BM25 table

`index/clip_nlp/ + nlp_docs FTS5`

QUERY · every request

Three candidate lists
CLIP · BGE doc · BM25 (1.0/1.6/1.8/0.6)
lexicon expansion enters BM25 only

Min-max fuse
0.5 CLIP + 0.3 doc + 0.2 BM25
empty signals drop out and renormalise

Rerank the top 60
cross-encoder on (query, document)
fused ← 0.7·fused + 0.3·rerank

Wins wherever the intent lives — metaphor 0.912, intent 0.739.

C · vlm

A VLM writes the annotation, offline

BUILD · once, offline

Caption the gallery offline
Qwen2.5-VL-3B → JSON per image
scene · OCR · meaning · metaphors

Embed the descriptions
doc_text → BGE-small, 384-d
descriptions.jsonl is the resume cache

`index/vlm/ · 384-d`

QUERY · every request

Encode the query
BGE with the query instruction
the VLM is never loaded here

Search and fuse
doc 0.8 + CLIP 0.2, min-max
no CLIP → the doc weight becomes 1.0

Return the ranking
no rerank by default
provenance checked before serving

Loses fine detail, recovers part of the intent — net ≈ A.

Comparison	A · clip	B · clip_nlp	C · vlm
What it matches	The pixels, nothing else.	Human annotation text + pixels.	A VLM's paraphrase + pixels.
Text per image	none	scene · meaning · metaphor · title	scene · OCR · meaning · metaphors
Indexes	index/clip · 6,378 × 768-d	+ index/clip_nlp · 384-d + FTS5	+ index/vlm · 384-d
Signals at query time	1 — CLIP text tower	3 — CLIP 0.5 · doc 0.3 · BM25 0.2	2 — doc 0.8 · CLIP 0.2
Post-processing	drop score ≤ 0, keep top k	cross-encoder on top 60, α = 0.3	none — rerank disabled
Candidates fetched	max(2k, k)	max(5k, 100) per signal	max(3k, 100) per signal
One-off cost · new memes	minutes · generalises	human annotation · no cold start	2 h 35 min GPU · generalises
Latency p50 / p95	14.4 / 20.8 ms	225.6 / 285.9 ms	35.3 / 41.6 ms
mAP on the frozen 240	0.4395	**0.7608**	0.4519

Weights from configs/default.yaml; behaviour read from strategies/*.py; metrics from data/reports/compare.json.

Offline build flow

Five commands turn the Meme-Cap download into the gallery and the three indexes. Every stage is keyed by a content hash, so a stale artefact is refused, never served.



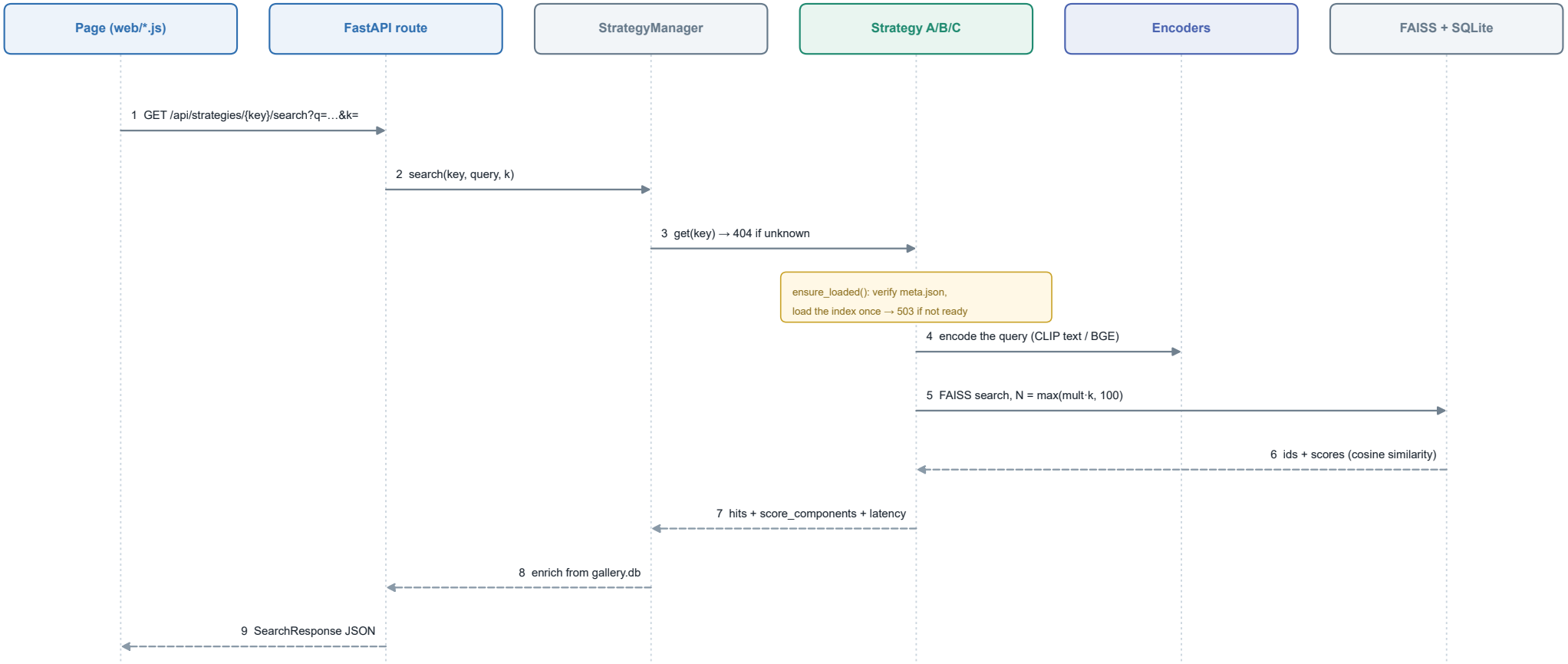
Dependency order: gallery before clip_nlp (it reuses the CLIP index) and before vlm (it requires vlm_docs). --strategy all runs clip → clip_nlp → vlm and skips C with a warning when models.vlm.backend is none. B and C never touch images.id: they only rewrite their own rows and vector files.

`python -m tsr stats --json` → readiness, counts and model strings for every stage

Command order and guard rails read from cli.py and pipeline.py; timings measured on a 12 GB RTX 4070 SUPER (captioning) and a desktop CPU (index builds).

Online query flow

One search request, from the page to the ranked payload. Models and indexes load on the first query of each strategy, never at process start.



Failure semantics

404 unknown strategy key
503 known, but the index is missing or stale
422 invalid settings payload
/compare keeps a failing column as {ready: false}

What loads when

startup config, SQLite, strategy objects
first query CLIP or BGE weights, FAISS files, reranker
never the VLM stack (captioning is offline)
an RLock serialises searches across threads

Measured latency · p50 / p95 over the 240 frozen queries

A · clip 14 / 21 ms
B · clip_nlp 226 / 286 ms
C · vlm 35 / 42 ms
bar = p50 · outline = p95 · reranking dominates B

Route order from api.py, readiness checks from service.py and strategies/vlm.py; latencies from data/reports/compare.json.

Evaluation and reporting flow

A frozen, leakage-checked benchmark is run through every ready strategy; the same report files drive the CLI and the /evaluation page.

1 BUILD THE BENCHMARK

Draft
from the 558 test images
80 visual · 80 intent · 80 metaphor

Rewrite
one reviewed rewrite per draft
data/eval/rewrites/*.json

Leakage gate
8-gram · whole short annotation
≥ 5-token copy · Jaccard ≥ 0.6

Freeze
memecap_eval_v1 · 240 queries
split and annotation snapshot checked

2 MEASURE EVERY STRATEGY

Same gallery
one image table, one CLIP index
all three strategies search it

Graded qrels
grade 2 = the source meme
grade 1 = deterministic rules only

Metrics
P@1/5/10 · R@1/5/10 · mAP · MRR
nDCG@10 · median rank · misses

Depth
max(top_k, largest relevant set, 100)
latency reported as p50 / p95

3 COMPARE AND PUBLISH

Paired bootstrap
1,000 resamples, seed 42
paired per-query differences

Correction
Holm–Bonferroni over
3 pairs × 5 metrics · p ≥ 0.001

Ablations
7 single-change variants
prompt ensemble · B channels · C fusion

Reports
compare.json · compare.html · failures.html
/api/eval/summary → /evaluation

LEAKAGE DIAGNOSTIC The untouched human annotations are also run as queries and reported separately (B +0.139 mAP, A −0.085, C −0.193). Leaky by construction — never mixed into the main table.

MEASURED RESULT · 240 frozen queries, identical gallery and retrieval depth

Strategy	mAP [95% CI]	MRR	nDCG@10	P@1	R@10	p50
A · clip	0.4395 [0.385, 0.498]	0.4461	0.4744	0.358	0.607	14 ms
B · clip_nlp	**0.7608 [0.713, 0.808]**	0.7647	0.7804	0.700	0.870	226 ms
C · vlm	0.4519 [0.399, 0.507]	0.4632	0.4881	0.383	0.626	35 ms

B − A mAP +0.321 [0.270, 0.371] p_holm = 0.015
B − C mAP +0.309 [0.255, 0.363] p_holm = 0.015
C − A mAP +0.012 [−0.038, 0.064] not significant

Per-category mAP (visual / intent / metaphor) · A 0.315 / 0.444 / 0.559 B 0.631 / 0.739 / 0.912 C 0.247 / 0.539 / 0.570

Ablations (mAP) · B − BM25 0.704 · B − dense 0.704 · B − CLIP 0.780 · B − reranker 0.689 · C pure VLM 0.392 · C + CLIP 0.35 0.472

Reading B wins by +0.32 mAP over both annotation-free strategies; C and A are statistically indistinguishable, so a 3B VLM is a cheap approximation of the annotation signal, not a substitute.

Sources: data/eval/memecap_eval_v1.{json,qrels.jsonl} and data/reports/compare.json (1,000 bootstrap resamples, seed 42).